

InverterLab

Uch fazali inverter tahlil hisoboti

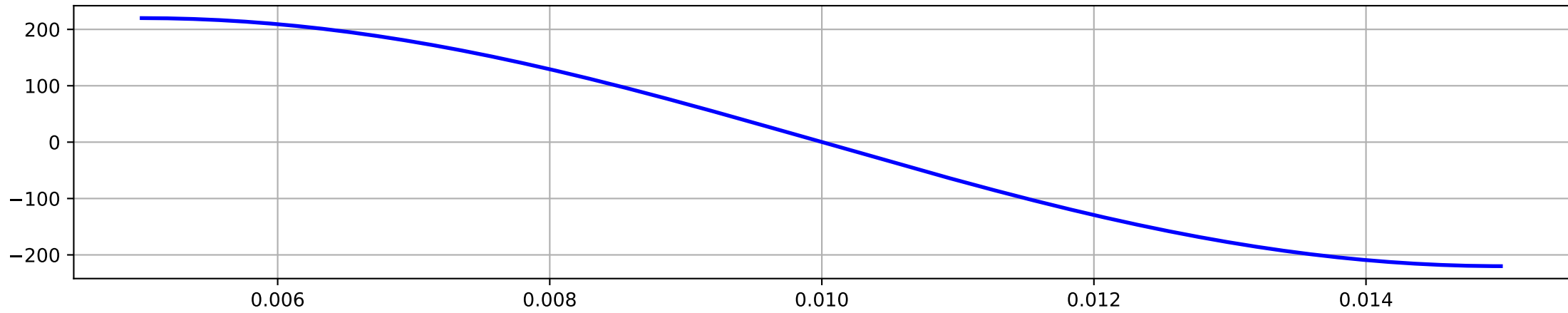
Muallif: Murodov Shohruh

Qo'qon Davlat Universiteti

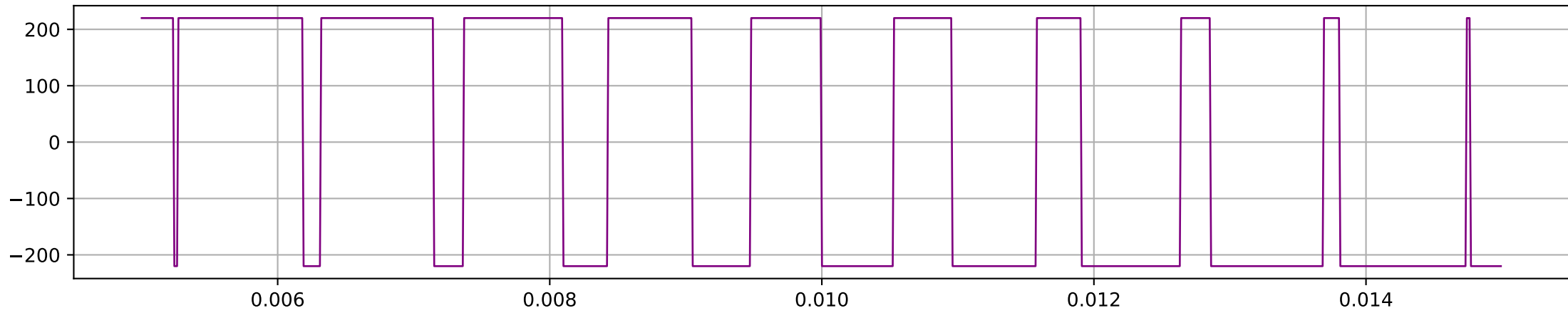
Sana: 11.05.2026

Signal tahlili

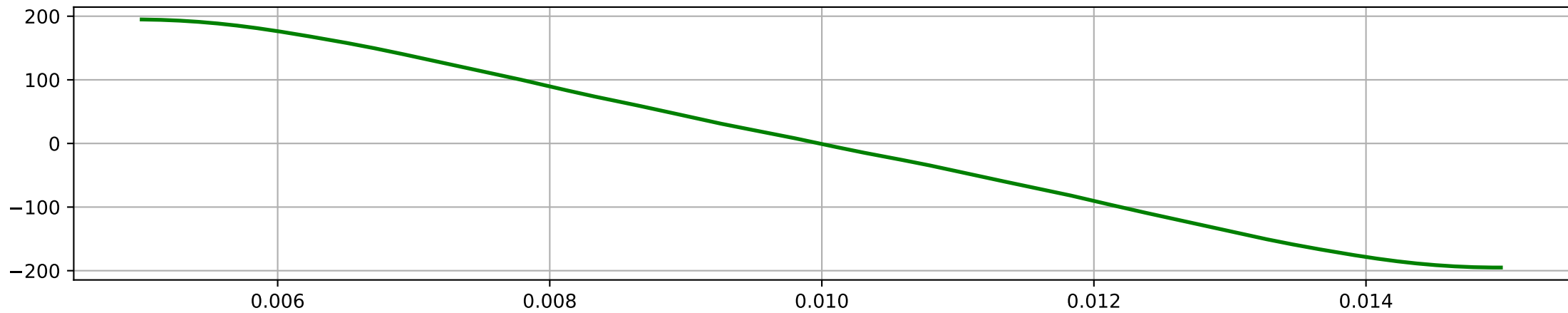
Ideal sinusoidal signal (220V, 50Hz)



PWM signal (THD=129.2%)

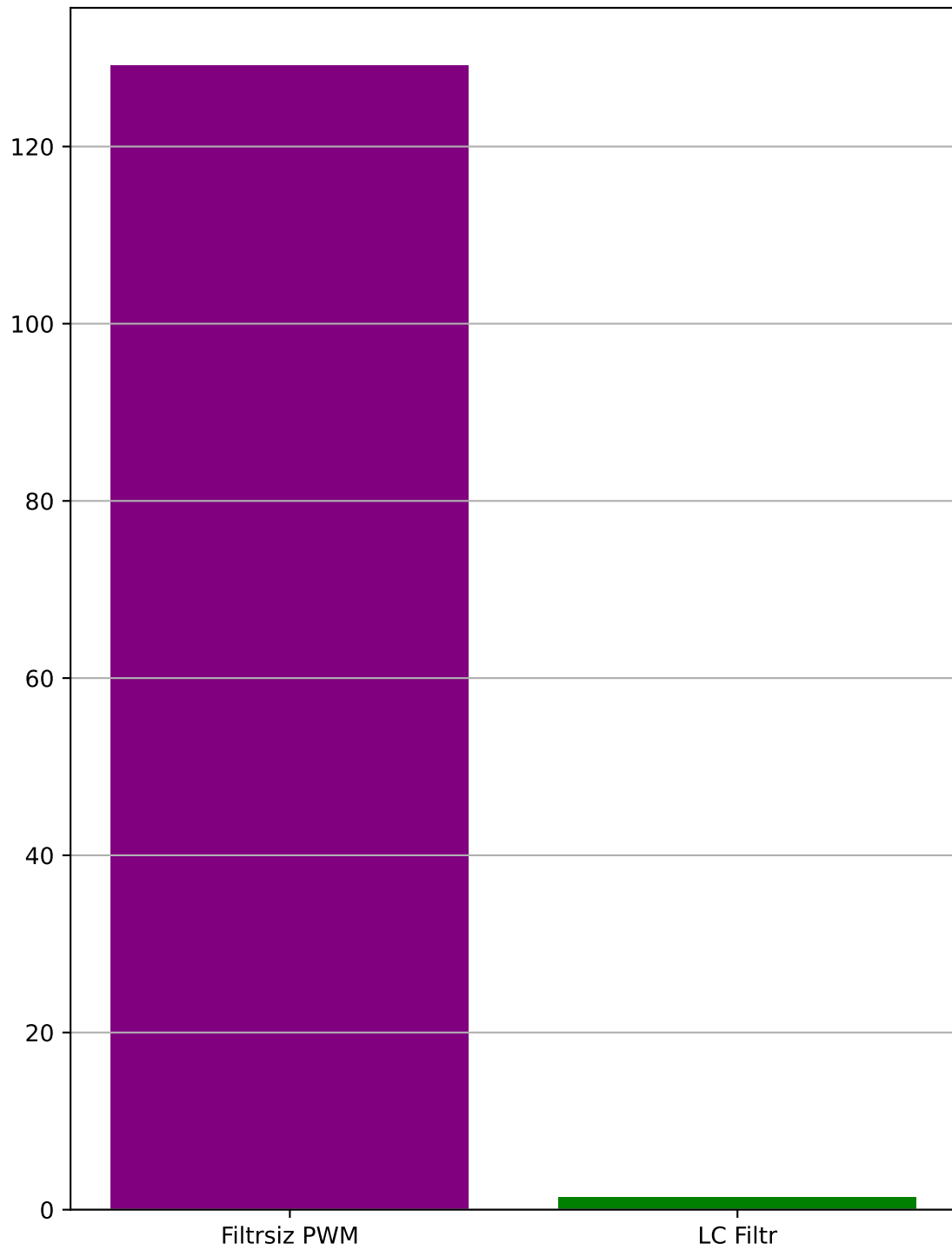


LC Filtr bilan (THD=1.4%)

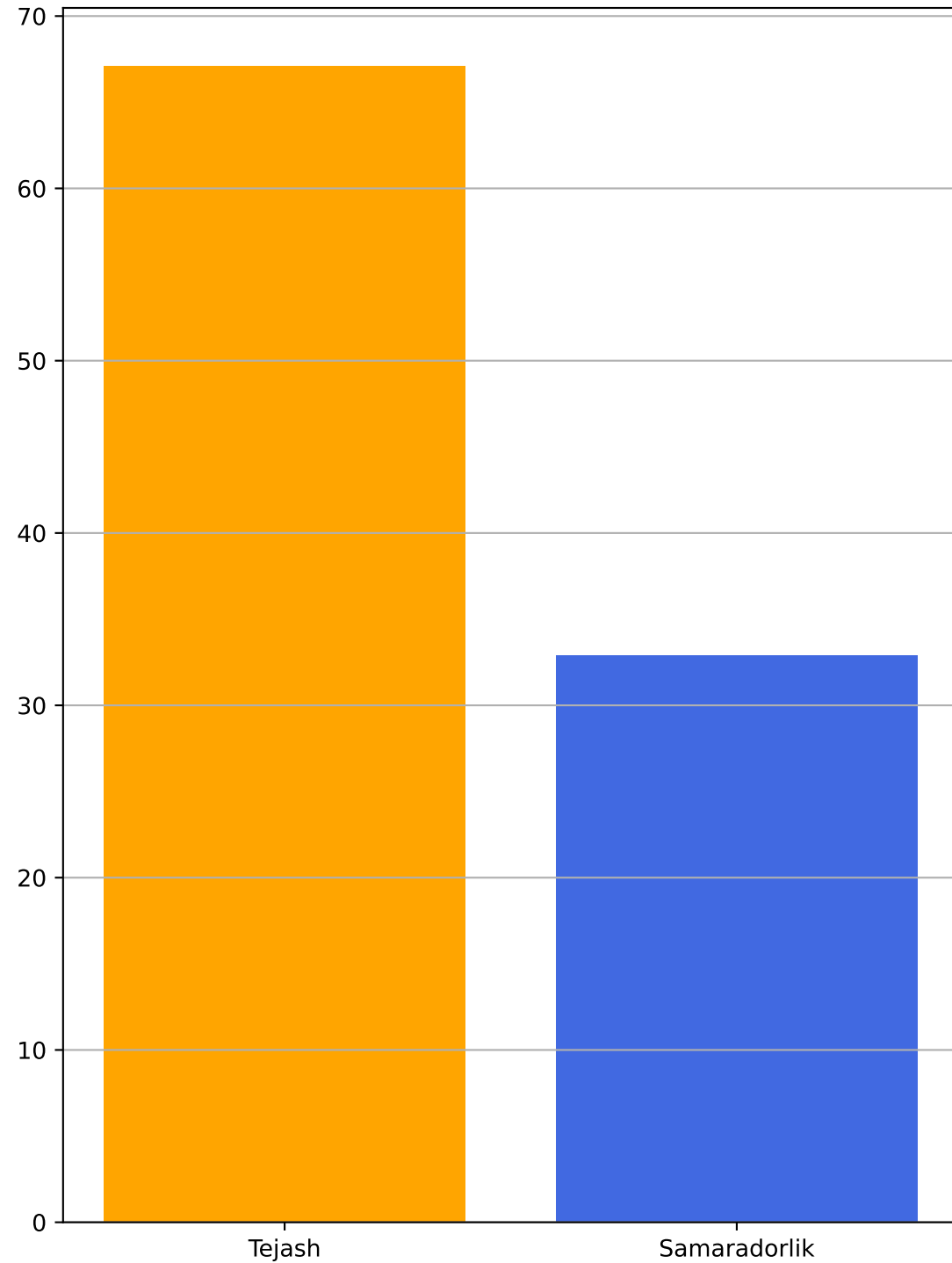


Natijalar

THD taqqoslash (%)



Energiya ko'rsatkichlari (%)



Xulosa

- Chastota: 50 Hz
- Kuchlanish: 220 V
- PWM chastotasi: 1000 Hz
- Filtrsiz THD: 129.2%
- Filtrlangan THD: 1.4%
- THD kamayishi: 127.8%
- Energiya tejash: 67.1%
- GitHub: github.com/MurodovShohruh/InverterLab